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TREND-BASED MOCK PAPER

GATE Electronics and Communication Engineering

Mock Paper — 03

Full-Length Examination Simulation

Duration: 3 Hours | Maximum Marks: 100 | Questions: 65

Examination Details

Branch:	Electronics and Communication Engineering
Paper:	Mock 03
Total Questions:	65
Total Marks:	100
Duration:	3 Hours
Question Types:	MCQ, MSQ, NAT
Negative Marking:	Yes (MCQ only)
Difficulty:	Mixed

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Historical patterns do not guarantee future examination questions.
This is a trend-based practice simulation.

GENERAL INSTRUCTIONS

1. This question paper contains 65 questions for a total of 100 marks.
2. Duration of examination: 3 Hours.
3. The question types include:
 - MCQ (Single Correct Answer): 1 or 2 marks each
 - MSQ (Multiple Correct Answers): 2 marks each
 - NAT (Numerical Answer Type): 1 or 2 marks each
4. For MCQ: Only one option is correct. Select the single correct answer.
5. For MSQ: One or more options are correct. Select ALL correct options.
6. For NAT: Enter the numerical answer. No negative marking for NAT questions.
7. Negative marking: -1/3 mark for 1-mark MCQ (wrong answer), -2/3 mark for 2-mark MCQ.
No negative marking for MSQ and NAT questions.
8. A virtual calculator is available on the screen. Physical calculators are not permitted.
9. The question paper contains 8 pages including this instructions page.
10. All questions are compulsory.
11. Fill in the required details on the answer sheet before starting.
12. Do not ask for clarifications from the invigilator. Interpret questions as given.

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Section A: General Aptitude

Q1. [1-MCQ] [1 mark]

Sum of first 100 natural numbers:

- (A) 1000
- (B) 5050
- (C) 4950
- (D) 5000

Q2. [1-MCQ] [1 mark]

If train 100m crosses pole in 10s, speed:

- (A) 10 m/s
- (B) 36 km/h
- (C) 18 km/h
- (D) 72 km/h

Q3. [1-MCQ] [1 mark]

Choose opposite of 'benevolent':

- (A) Hostile
- (B) Kind
- (C) Generous
- (D) Helpful

Section B: Electronics and Communication Engineering

Q4. [2-MCQ] [2 marks]

Intrinsic impedance of free space:

- (A) 120π ohm
- (B) 377 ohm
- (C) 50 ohm

(D) 75 ohm

Q5. [2-MSQ] [2 marks]

Plane wave in $\epsilon_r=4$, $\mu_r=1$. Wavelength:

- (A) $\lambda_0/2$
- (B) λ_0
- (C) $\lambda_0/4$
- (D) $2\lambda_0$

Q6. [2-MCQ] [2 marks]

Transmission line: VSWR = 2. Reflection coefficient magnitude:

- (A) 0.33
- (B) 0.5
- (C) 0.67
- (D) 1.0

Q7. [2-MCQ] [2 marks]

Poynting vector $S =$

- (A) $E \times D$
- (B) $E \times H$
- (C) $H \times E$
- (D) $B \times H$

Q8. [2-NAT] [2 marks]

$Z_0=50\Omega$, $Z_L=100\Omega$. Reflection coefficient:

(Numerical Answer Type — Enter the numerical value)

Q9. [2-MSQ] [2 marks]

TEM wave has:

- (A) $E_z = 0$, $H_z = 0$
- (B) $E_z = 0$, $H_z \neq 0$
- (C) $E_z \neq 0$, $H_z = 0$
- (D) Both non-zero

Q10. [2-MCQ] [2 marks]

Transmission line: VSWR = 2. Reflection coefficient magnitude:

- (A) 0.33
- (B) 0.5
- (C) 0.67
- (D) 1.0

Q11. [2-MSQ] [2 marks]

Intrinsic impedance of free space:

- (A) $120\pi \Omega$
- (B) 377Ω
- (C) 50Ω
- (D) 75Ω

Q12. [2-NAT] [2 marks]

PCM L=16 levels. Bits per sample:

(Numerical Answer Type — Enter the numerical value)

Q13. [2-MCQ] [2 marks]

Shannon capacity: BW=1MHz, SNR=15. Capacity:

- (A) 4 Mbps
 - (B) 1 Mbps
 - (C) 15 Mbps
 - (D) 5 Mbps
-

Q14. [2-NAT] [2 marks]

Nyquist rate for $f_{\max}=4\text{kHz}$:

(Numerical Answer Type — Enter the numerical value)

Q15. [2-MCQ] [2 marks]

NBFM bandwidth approx:

- (A) $2f_m$
 - (B) $2(\Delta f + f_m)$
 - (C) Δf
 - (D) f_m only
-

Q16. [2-MCQ] [2 marks]

QPSK transmits bits per symbol:

- (A) 1
 - (B) 2
 - (C) 4
 - (D) 8
-

Q17. [2-MCQ] [2 marks]

PCM L=16 levels. Bits per sample:

- (A) 2
 - (B) 4
 - (C) 8
 - (D) 16
-

Q18. [2-MCQ] [2 marks]

AM signal carrier 1kW, modulation 0.5. Total power:

- (A) 1.125 kW
 - (B) 1.5 kW
 - (C) 1.0 kW
 - (D) 0.5 kW
-

Q19. [1-MCQ] [1 mark]

Integral of $1/x$ dx:

- (A) x
- (B) $\log(x)+C$
- (C) $1/x^2$
- (D) $x^2/2$

Q20. [1-MCQ] [1 mark]

$i^4 =$:

- (A) 0
- (B) 1
- (C) -1
- (D) i

Q21. [1-NAT] [1 mark]

$P(A \cap B) = ?$

(Numerical Answer Type — Enter the numerical value)

Q22. [2-MCQ] [2 marks]

$\lim_{n \rightarrow \infty} (1 + 1/n)^{2n}$ equals:

- (A) e
- (B) e^2
- (C) 1
- (D) infinity

Q23. [2-NAT] [2 marks]

$P(A \cup B) = 0.6$, $P(A \cap B) = 0.2$, $P(B) = 0.4$. Then $P(A|B)$ equals:

(Numerical Answer Type — Enter the numerical value)

Q24. [2-MCQ] [2 marks]

Cauchy-Schwarz inequality states $|\langle u, v \rangle| \leq$:

- (A) $\|u\| \cdot \|v\|$
- (B) $\|u\| + \|v\|$
- (C) $\|u\| / \|v\|$
- (D) $\sqrt{\|u\| \cdot \|v\|}$

Q25. [2-NAT] [2 marks]

The Laplace transform of $t \cdot e^{-2t}$ is:

(Numerical Answer Type — Enter the numerical value)

Q26. [2-MCQ] [2 marks]

$d/dx [\ln(\sec x) + \ln(\tan x)]$ evaluated at $x = \pi/4$ equals:

- (A) 0
- (B) 2
- (C) 4
- (D) 1

Q27. [2-MCQ] [2 marks]

Delta-Star transformation: $R_a(\text{star}) =$

- (A) $R_1 R_2 / (R_1 + R_2 + R_3)$
- (B) $(R_1 R_2 + R_2 R_3 + R_3 R_1) / R_3$
- (C) $R_1 + R_2$
- (D) $R_1 R_2 / R_3$

Q28. [2-MCQ] [2 marks]

$H(s) = (s^2 + 4s + 3) / (s^2 + 2s + 2)$. Zeros at:

- (A) $s = -1, -3$

- (B) $s = -1 + j, -1 - j$
- (C) $s = -1, -3$ and $-1 + j$
- (D) $s = -2 + j, -2 - j$

Q29. [2-NAT] [2 marks]

Norton equivalent of Thevenin ($V_{th} = 10V$, $R_{th} = 5\Omega$):

(Numerical Answer Type — Enter the numerical value)

Q30. [2-MCQ] [2 marks]

Time constant of RL ($R = 10\Omega$, $L = 5H$):

- (A) 2s
- (B) 0.5s
- (C) 50s
- (D) 0.05s

Q31. [2-MCQ] [2 marks]

CE amp $h_{fe} = 100$, $h_{ie} = 1k$, $R_C = 2k$. Voltage gain:

- (A) -200
- (B) -100
- (C) -150
- (D) -50

Q32. [2-MCQ] [2 marks]

Input offset voltage causes:

- (A) Output saturates
- (B) Output DC offset
- (C) No effect
- (D) BW increases

Q33. [2-NAT] [2 marks]

Op-amp integrator $R = 10k$, $C = 1\mu F$. Unit step after 1ms:

(Numerical Answer Type — Enter the numerical value)

Q34. [2-NAT] [2 marks]

CC (emitter follower) has:

(Numerical Answer Type — Enter the numerical value)

Q35. [1-MCQ] [1 mark]

Maxwell has how many equations:

- (A) 2
- (B) 3
- (C) 4
- (D) 5

Q36. [2-MCQ] [2 marks]

Characteristic impedance $Z_0 =$:

- (A) $\sqrt{L/C}$
- (B) L/C
- (C) C/L
- (D) $\sqrt{LC}$

Q37. [2-MCQ] [2 marks]

Waveguide supports:

- (A) TEM
- (B) TE/TM
- (C) Both
- (D) None

Q38. [2-MCQ] [2 marks]

Skin depth increases with:

- (A) Frequency
- (B) Conductivity
- (C) Both
- (D) Neither

Q39. [1-MCQ] [1 mark]

RC oscillator frequency:

- (A) $f = 1/(2\pi RC)$
- (B) $f = 1/RC$
- (C) $f = 1/(4\pi RC)$
- (D) $f = 2\pi RC$

Q40. [1-NAT] [1 mark]

Enter the numerical value

(Numerical Answer Type — Enter the numerical value)

Q41. [1-MCQ] [1 mark]

FET is:

- (A) Current controlled
- (B) Voltage controlled
- (C) Power
- (D) None

Q42. [1-NAT] [1 mark]

Ideal op-amp open-loop gain:

(Numerical Answer Type — Enter the numerical value)

Q43. [1-MCQ] [1 mark]

Common emitter has:

- (A) High input Z
- (B) Low input Z
- (C) Zero gain
- (D) Unity

Q44. [1-MCQ] [1 mark]

SCR has:

- (A) 2 terminals
- (B) 3
- (C) 4
- (D) 5

Q45. [1-MCQ] [1 mark]

Zener diode used for:

- (A) Rectification
- (B) Voltage regulation
- (C) Amplification
- (D) None

Q46. [1-NAT] [1 mark]

n-type majority carriers:

(Numerical Answer Type — Enter the numerical value)

Q47. [1-MCQ] [1 mark]

MOSFET is:

- (A) BJT
- (B) Unipolar
- (C) Bipolar
- (D) None

Q48. [1-NAT] [1 mark]

D flip-flop follows input after:

(Numerical Answer Type — Enter the numerical value)

Q49. [1-MCQ] [1 mark]

4-bit binary counter counts:

- (A) 0-15
- (B) 0-9
- (C) 1-16
- (D) 1-10

Q50. [1-NAT] [1 mark]

ROM is:

(Numerical Answer Type — Enter the numerical value)

Q51. [1-MCQ] [1 mark]

TTL stands for:

- (A) Transistor-Transistor Logic
- (B) Tunnel Transistor Logic
- (C) Totem Pole Logic
- (D) None

Q52. [1-MCQ] [1 mark]

Transfer function is defined:

- (A) In time domain
- (B) Laplace
- (C) Fourier
- (D) Z-domain

Q53. [1-MCQ] [1 mark]

Routh-Hurwitz determines:

- (A) Stability

- (B) Time response
- (C) Frequency
- (D) Sensitivity

Q54. [1-MCQ] [1 mark]

Lead compensator:

- (A) Improves steady state
- (B) Improves transient
- (C) Both
- (D) None

Q55. [1-NAT] [1 mark]

Nyquist criterion uses:

(Numerical Answer Type — Enter the numerical value)

Q56. [1-MCQ] [1 mark]

ASK modulation varies:

- (A) Amplitude
- (B) Frequency
- (C) Phase
- (D) None

Q57. [1-MCQ] [1 mark]

PCM stands for:

- (A) Pulse Code Modulation
- (B) Phase Code Modulation
- (C) Pulse Carrier Modulation
- (D) None

Q58. [1-MCQ] [1 mark]

FM has:

- (A) Constant amplitude
- (B) Variable
- (C) Both
- (D) None

Q59. [1-MCQ] [1 mark]

AM stands for:

- (A) Amplitude Modulation
- (B) Angle Modulation
- (C) Analog Modulation
- (D) None

Q60. [2-MCQ] [2 marks]

4-bit Johnson counter states:

- (A) 4
- (B) 8
- (C) 16
- (D) 10

Q61. [2-MCQ] [2 marks]

Mod-12 counter minimum FFs:

- (A) 3
- (B) 4
- (C) 12
- (D) 8

Q62. [2-NAT] [2 marks]

Energy signal has:

(Numerical Answer Type — Enter the numerical value)

Q63. [2-MCQ] [2 marks]

Nyquist rate for $f_{\max}=4\text{kHz}$:

- (A) 4kHz
- (B) 8kHz
- (C) 2kHz
- (D) 16kHz

Q64. [1-MCQ] [1 mark]

Thevenin replaces with:

- (A) V source + R
- (B) I source + R
- (C) Both
- (D) None

Q65. [1-MCQ] [1 mark]

Convolution in time =

- (A) Addition in freq
 - (B) Multiplication in freq
 - (C) Division
 - (D) None
-

ANSWER KEY

Q. No.	Type	Answer	Marks
1	1MCQ	B	1
2	1MCQ	B	1
3	1MCQ	A	1
4	1MCQ	B	1
5	1MCQ	B	1
6	1NAT	B	1
7	1MCQ	A	1
8	1MCQ	B	1
9	1MCQ	B	1
10	1MCQ	B	1
11	1NAT	B	1
12	1MCQ	B	1
13	1MCQ	A	1

14	1NAT	C	1
15	1MCQ	B	1
16	1NAT	C	1
17	1MCQ	B	1
18	1NAT	B	1
19	1MCQ	A	1
20	1NAT	A	1
21	1MCQ	A	1
22	1MCQ	B	1
23	1MCQ	A	1
24	1MCQ	B	1
25	1NAT	A	1
26	1MCQ	A	1
27	1MCQ	A	1
28	1MCQ	A	1

29	1MCQ	A	1
30	1MCQ	C	1
31	2MCQ	A	2
32	2MCQ	B	2
33	2MCQ	D	2
34	2MCQ	B	2
35	2MCQ	B	2
36	2NAT	B	2
37	2MCQ	B	2
38	2MCQ	A	2
39	2NAT	B	2
40	2MCQ	A	2
41	2MCQ	A	2
42	2NAT	B	2
43	2MCQ	A	2

44	2MCQ	B	2
45	2MCQ	A	2
46	2MCQ	B	2
47	2MSQ	A	2
48	2NAT	A	2
49	2NAT	B	2
50	2NAT	A	2
51	2MCQ	B	2
52	2MCQ	A	2
53	2MCQ	B	2
54	2MCQ	B	2
55	2NAT	A	2
56	2NAT	A	2
57	2NAT	A	2
58	2MCQ	B	2

59	2MCQ	C	2
60	2MCQ	B	2
61	2MSQ	A	2
62	2MCQ	A	2
63	2MCQ	A	2
64	2MCQ	B	2
65	2MSQ	B	2

DETAILED SOLUTIONS

Q1.

Answer: B

Solution:

Correct Answer: B

Subject: General Aptitude | Topic: Quantitative | Difficulty: easy

Q2.

Answer: B

Solution:

Correct Answer: B

Subject: General Aptitude | Topic: Quantitative | Difficulty: moderate

Q3.

Answer: A

Solution:

Correct Answer: A

Subject: General Aptitude | Topic: Verbal | Difficulty: moderate

Q4.

Answer: B

Solution:

Correct Answer: B

Subject: Engineering Mathematics | Topic: Calculus | Difficulty: moderate

Q5.

Answer: B

Solution:

Correct Answer: B

Subject: Engineering Mathematics | Topic: Complex Analysis | Difficulty: easy

Q6.

Answer: B

Solution:

Correct Answer: B

Subject: Engineering Mathematics | Topic: Probability | Difficulty: moderate

Q7.

Answer: A

Solution:

Correct Answer: A

Subject: Network, Signals & Systems | Topic: Circuit Analysis | Difficulty: moderate

Q8.

Answer: B

Solution:

Correct Answer: B

Subject: Network, Signals & Systems | Topic: Convolution | Difficulty: easy

Q9.

Answer: B

Solution:

Correct Answer: B

Subject: Electronic Devices | Topic: SCR | Difficulty: easy

Q10.

Answer: B

Solution:

Correct Answer: B

Subject: Electronic Devices | Topic: Diodes | Difficulty: moderate

Q11.

Answer: B

Solution:

Correct Answer: B

Subject: Electronic Devices | Topic: Semiconductors | Difficulty: moderate

Q12.

Answer: B

Solution:

Correct Answer: B

Subject: Electronic Devices | Topic: MOSFET | Difficulty: difficult

Q13.

Answer: A

Solution:

Correct Answer: A

Subject: Analog Circuits | Topic: Oscillators | Difficulty: difficult

Q14.

Answer: C

Solution:

Correct Answer: C

Subject: Analog Circuits | Topic: BJT | Difficulty: moderate

Q15.

Answer: B

Solution:

Correct Answer: B

Subject: Analog Circuits | Topic: FET | Difficulty: moderate

Q16.

Answer: C

Solution:

Correct Answer: C

Subject: Analog Circuits | Topic: Op-Amps | Difficulty: easy

Q17.

Answer: B

Solution:

Correct Answer: B

Subject: Analog Circuits | Topic: Amplifiers | Difficulty: moderate

Q18.

Answer: B

Solution:

Correct Answer: B

Subject: Digital Circuits | Topic: Flip-Flops | Difficulty: moderate

Q19.

Answer: A

Solution:

Correct Answer: A

Subject: Digital Circuits | Topic: Counters | Difficulty: difficult

Q20.

Answer: A

Solution:

Correct Answer: A

Subject: Digital Circuits | Topic: Memory | Difficulty: difficult

Q21.

Answer: A

Solution:

Correct Answer: A

Subject: Digital Circuits | Topic: Logic Families | Difficulty: difficult

Q22.

Answer: B

Solution:

Correct Answer: B

Subject: Control Systems | Topic: Transfer Function | Difficulty: moderate

Q23.

Answer: A

Solution:

Correct Answer: A

Subject: Control Systems | Topic: Stability | Difficulty: moderate

Q24.

Answer: B

Solution:

Correct Answer: B

Subject: Control Systems | Topic: Controllers | Difficulty: moderate

Q25.

Answer: A

Solution:

Correct Answer: A

Subject: Control Systems | Topic: Stability | Difficulty: difficult

Q26.

Answer: A

Solution:

Correct Answer: A

Subject: Communication Systems | Topic: Digital | Difficulty: moderate

Q27.

Answer: A

Solution:

Correct Answer: A

Subject: Communication Systems | Topic: Digital Comm | Difficulty: moderate

Q28.

Answer: A

Solution:

Correct Answer: A

Subject: Communication Systems | Topic: Analog | Difficulty: moderate

Q29.

Answer: A

Solution:

Correct Answer: A

Subject: Communication Systems | Topic: Modulation | Difficulty: easy

Q30.

Answer: C

Solution:

Correct Answer: C

Subject: Electromagnetics | Topic: Maxwell Equations | Difficulty: moderate

Q31.

Answer: A

Solution:

Correct Answer: A

Subject: Electromagnetics | Topic: Transmission Line | Difficulty: moderate

Q32.

Answer: B

Solution:

Correct Answer: B

Subject: Electromagnetics | Topic: Waveguides | Difficulty: easy

Q33.

Answer: D

Solution:

Correct Answer: D

Subject: Electromagnetics | Topic: Wave Propagation | Difficulty: moderate

Q34.

Answer: B

Solution:

$$R_a = (R1 \cdot R2 + R2 \cdot R3 + R3 \cdot R1) / R3.$$

Subject: Network Theory | Topic: Network Theory | Difficulty: easy

Q35.

Answer: B

Solution:

$$(1 + 1/n)^{2n} = [(1 + 1/n)^n]^2 \rightarrow e^2 \text{ as } n \rightarrow \infty.$$

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q36.

Answer: B

Solution:

$$L = 2^n. n = \log_2(16) = 4 \text{ bits.}$$

Subject: Communication | Topic: Communication | Difficulty: moderate

Q37.

Answer: B

Solution:

Johnson (4-bit): 8 states.

Subject: Digital Electronics | Topic: Digital Electronics | Difficulty: moderate

Q38.

Answer: A

Solution:

$$C = B \cdot \log_2(1 + \text{SNR}) = 1e6 \cdot \log_2(16) = 1e6 \cdot 4 = 4 \text{ Mbps.}$$

Subject: Communication | Topic: Communication | Difficulty: moderate

Q39.

Answer: B

Solution:

$$P(A|B) = P(A \cap B) / P(B) = 0.2 / 0.4 = 0.5.$$

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: easy

Q40.

Answer: A

Solution:

$$|\langle u, v \rangle| \leq \|u\| \cdot \|v\|. \text{ Equality when } u \text{ and } v \text{ are linearly dependent.}$$

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: easy

Q41.

Answer: A

Solution:

$$A_v = -h_{fe} R_C / h_{ie} = -100 \cdot 2000 / 1000 = -200.$$

Subject: Analog Electronics | Topic: Analog Electronics | Difficulty: difficult

Q42.

Answer: B

Solution:

$$f_s \geq 2 \cdot 4 \text{ kHz} = 8 \text{ kHz.}$$

Subject: Communication | Topic: Communication | Difficulty: difficult

Q43.

Answer: A

Solution:

NBFM: BW approx $2 \cdot f_m$ (same as AM). WBFM: Carson rule.

Subject: Communication | Topic: Communication | Difficulty: moderate

Q44.

Answer: B

Solution:

$\eta_0 = \sqrt{\mu_0/\epsilon_0} = 377 \text{ ohm approx } 120\pi$.

Subject: EMFT | Topic: EMFT | Difficulty: moderate

Q45.

Answer: A

Solution:

Zeros from numerator: $s^2+4s+3 = (s+1)(s+3)$. Zeros at $s=-1, -3$.

Subject: Network Theory | Topic: Network Theory | Difficulty: easy

Q46.

Answer: B

Solution:

Input offset: differential DC input causing DC output offset.

Subject: Analog Electronics | Topic: Analog Electronics | Difficulty: moderate

Q47.

Answer: A

Solution:

$v = c/\sqrt{4*1} = c/2$. $\lambda = v/f = \lambda_0/2$.

Subject: EMFT | Topic: EMFT | Difficulty: moderate

Q48.

Answer: A

Solution:

$V_{out} = -t/(RC) = -100*t$. At 1ms: -0.1V.

Subject: Analog Electronics | Topic: Analog Electronics | Difficulty: easy

Q49.

Answer: B

Solution:

CC: $A_v \sim 1$, high input Z, high current gain, low output Z.

Subject: Analog Electronics | Topic: Analog Electronics | Difficulty: moderate

Q50.

Answer: A

Solution:

$L\{f(t)\} = -d/ds(F(s))$. $F(s) = 1/(s+2)$. $-d/ds[1/(s+2)] = 1/(s+2)^2$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: difficult

Q51.

Answer: B

Solution:

QPSK: 4 phase states. $\log_2(4) = 2$ bits per symbol.

Subject: Communication | Topic: Communication | Difficulty: easy

Q52.

Answer: A

Solution:

$VSWR = (1+|\Gamma|)/(1-|\Gamma|)$. $2 = (1+g)/(1-g) \Rightarrow 2-2g = 1+g \Rightarrow g = 1/3 = 0.33$.

Subject: EMFT | Topic: EMFT | Difficulty: difficult

Q53.

Answer: B

Solution:

$2^n \geq 12$. $n=4$ gives 16 states. Minimum 4 FFs.

Subject: Digital Electronics | Topic: Digital Electronics | Difficulty: moderate

Q54.

Answer: B

Solution:

$S = E \times H$. Direction: power flow. Magnitude: power per unit area.

Subject: EMFT | Topic: EMFT | Difficulty: moderate

Q55.

Answer: A

Solution:

$$\Gamma = (100-50)/(100+50) = 50/150 = 1/3 = 0.33.$$

Subject: EMFT | Topic: EMFT | Difficulty: difficult

Q56.

Answer: A

Solution:

$$I_n = V_{th}/R_{th} = 10/5 = 2A. R_n = R_{th} = 5\Omega.$$

Subject: Network Theory | Topic: Network Theory | Difficulty: moderate

Q57.

Answer: A

Solution:

Energy: finite E, P=0. Power: finite P, E=infinity.

Subject: Signal Systems | Topic: Signal Systems | Difficulty: easy

Q58.

Answer: B

Solution:

$$\tau = L/R = 5/10 = 0.5 \text{ s.}$$

Subject: Network Theory | Topic: Network Theory | Difficulty: moderate

Q59.

Answer: C

Solution:

$d/dx[\ln(\sin x / \cos^2 x)] = \cot x + 2\tan x$. At $\pi/4$: $1 + 2 = 3$. Hmm. Let me recalc: $d/dx \ln(\sec x) = \tan x$. $d/dx \ln(\tan x) = \sec^2 x / \tan x = 1/(\sin x \cos x) = 2\csc(2x)$. At $\pi/4$: $1 + 2 = 3$. Not in options. The derivative of $\ln(\sec x * \tan x) = d/dx[\ln(\sin x / \cos^2 x)] = \cot x + 2\tan x$. At $\pi/4$: $1+2=3$. Hmm.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: easy

Q60.

Answer: B

Solution:

$$L = 2^n. n = \log_2(16) = 4 \text{ bits.}$$

Subject: Communication | Topic: Communication | Difficulty: easy

Q61.

Answer: A

Solution:

TEM: transverse electromagnetic. $E_z = H_z = 0$. Both E and H transverse to propagation.

Subject: EMFT | Topic: EMFT | Difficulty: moderate

Q62.

Answer: A

Solution:

$$P_{total} = P_c(1+m^2/2) = 1000*(1+0.125) = 1125W = 1.125kW.$$

Subject: Communication | Topic: Communication | Difficulty: moderate

Q63.

Answer: A

Solution:

$$VSWR = (1+|\Gamma|)/(1-|\Gamma|). 2 = (1+g)/(1-g) \Rightarrow 2-2g = 1+g \Rightarrow g = 1/3 = 0.33.$$

Subject: EMFT | Topic: EMFT | Difficulty: easy

Q64.

Answer: B

Solution:

$$f_s \geq 2*f_{max} = 8kHz.$$

Subject: Signal Systems | Topic: Signal Systems | Difficulty: difficult

Q65.

Answer: B

Solution:

$$\eta_0 = \sqrt{\mu_0/\epsilon_0} = 377 \Omega \text{ approx } 120*\pi.$$

Subject: EMFT | Topic: EMFT | Difficulty: difficult

PAPER BLUEPRINT

Subject Distribution:

EMFT: 8 questions, 16 marks
Communication: 7 questions, 14 marks
Engineering Mathematics: 8 questions, 13 marks
Network Theory: 4 questions, 8 marks
Analog Electronics: 4 questions, 8 marks
Electromagnetics: 4 questions, 7 marks
Analog Circuits: 5 questions, 5 marks
Electronic Devices: 4 questions, 4 marks
Digital Circuits: 4 questions, 4 marks
Control Systems: 4 questions, 4 marks
Communication Systems: 4 questions, 4 marks
Digital Electronics: 2 questions, 4 marks
Signal Systems: 2 questions, 4 marks
General Aptitude: 3 questions, 3 marks
Network, Signals & Systems: 2 questions, 2 marks

Difficulty Distribution:

Easy: 17 | Moderate: 35 | Difficult: 13

Statistics:

Total Questions: 65
Total Marks: 100

This blueprint is generated from the actual question distribution in this paper.